

SAE

J1939-71 REVISED JUN2006

- 28 -

SPN 114 **Net Battery Current**

Net flow of electrical current into/out of the battery or batteries.

Data Length: 1 byte

Resolution: 1 A/bit, -125 A offset

Data Range: -125 to 125 A

Operational Range: same as data range

Type: Measured

Supporting information:

PGN reference: 65271

SPN 168 **Electrical Potential (Voltage)**

Measured electrical potential of the battery.

Data Length: 2 bytes

Resolution: 0.05 V/bit, 0 offset

Data Range: 0 to 3212.75 V

Operational Range: same as data range

Type: Measured

Supporting information:

PGN reference: 65271

SPN 1800 **Battery 1 Temperature**

Temperature of the battery 1. The relation to physical location is determined by the equipment manufacturer.

Data Length: 1 byte

Resolution: 1 deg C/bit, -40 deg C offset

Data Range: -40 to 210 deg C

Operational Range: same as data range

Type: Measured

Supporting information:

PGN reference: 65104

SPN 2579 **Net Battery Current (High Range/Resolution)**

Net flow of electrical current into/out-of the battery or batteries. This parameter is the high range and resolution of SPN 114 - Net Battery Current.

Data Length: 2 bytes

Resolution: 0.05 A/bit, -1600 A offset

Data Range: -1600 to 1612.75 A

Operational Range: same as data range

Type: Measured

Supporting information:

PGN reference: 65106

PGN 65104**Battery Temperature****- BT1**

Contains battery temperature information.

Transmission Repetition Rate: 1 s
 Data Length: 8
 Extended Data Page: 0
 Data Page: 0
 PDU Format: 254
 PDU Specific: 80 PGN Supporting Information: See Appendix D - PGN 65104
 Default Priority: 6
 Parameter Group Number: 65104 (0xFE50)

Start Position	Length	Parameter Name	SPN
1	1 byte	Battery 1 Temperature	1800
2	1 byte	Battery 2 Temperature	1801

PGN 65106**Vehicle Electrical Power #3****- VEP3**

This contains high resolution/range parameters reported from the alternator or power generation components.

Transmission Repetition Rate: 1 s
 Data Length: 8
 Extended Data Page: 0
 Data Page: 0
 PDU Format: 254
 PDU Specific: 82 PGN Supporting Information: See Appendix D - PGN 65104
 Default Priority: 6
 Parameter Group Number: 65106 (0xFE52)

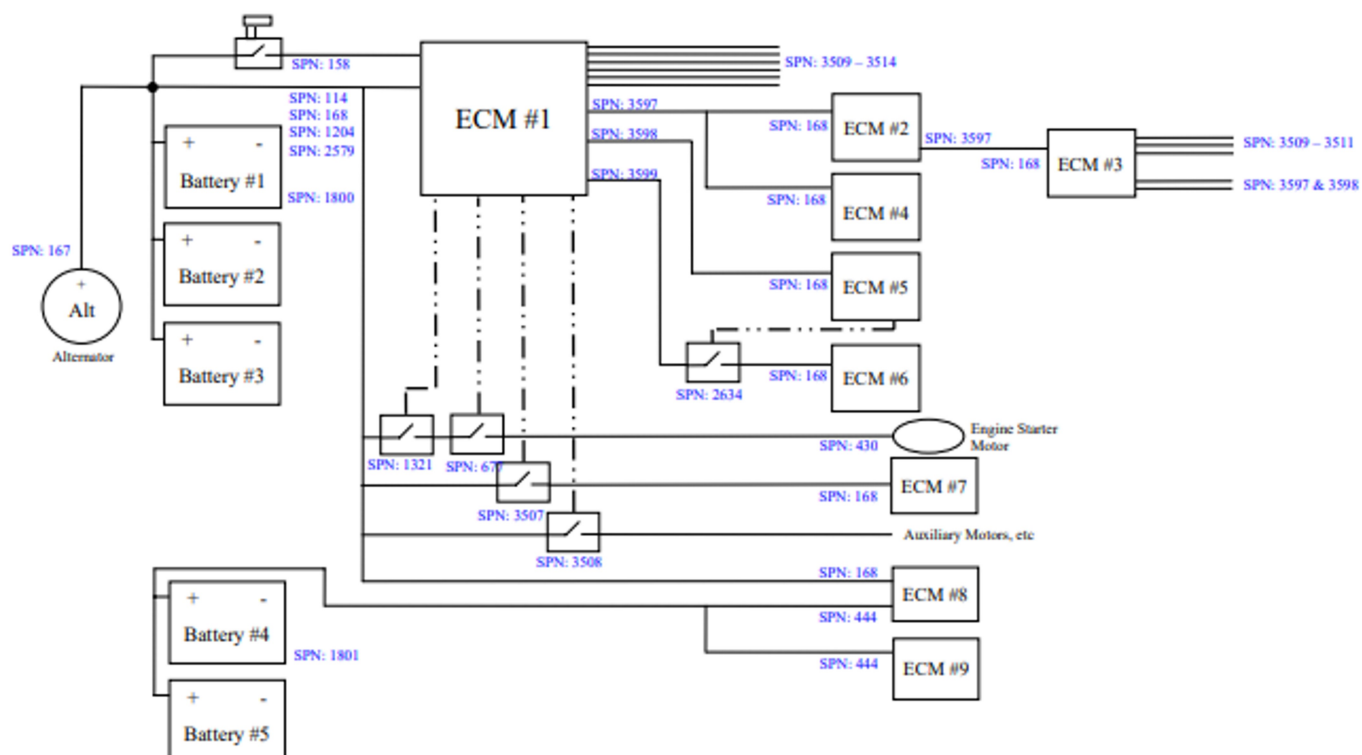
Start Position	Length	Parameter Name	SPN
1-2	2 bytes	Alternator Current (High Range/Resolution)	1795
3-4	2 bytes	Net Battery Current (High Range/Resolution)	2579

PGN 65271**Vehicle Electrical Power #1****- VEP1**

Transmission Repetition Rate: 1 s
 Data Length: 8
 Extended Data Page: 0
 Data Page: 0
 PDU Format: 254
 PDU Specific: 247 PGN Supporting Information:
 Default Priority: 6
 Parameter Group Number: 65271 (0xFE7)

Start Position	Length	Parameter Name	SPN
1	1 byte	Net Battery Current	114
2	1 byte	Alternator Current	115
3-4	2 bytes	Charging System Potential (Voltage)	167
5-6	2 bytes	Electrical Potential (Voltage)	168
7-8	2 bytes	Battery Potential (Voltage), Switched	158

(R) PGN 65104 – BATTERY INFORMATION



Note: The ECMs shown in this diagram could represent an ECM, or any intelligent device that is capable of diagnostics.

(R) FIGURE PGN65104_A – BATTERY INFORMATION

SPN	SPN Name	SPN	SPN Name
114	Net Battery Current	2579	Net Battery Current (High Range/Resolution)
115	Alternator Current	2634	Power Relay
158	Keyswitch Battery Potential	3507	TECU ECU PWR Relay
167	Charging System Potential (Voltage)	3508	TECU PWR Relay
168	Battery Potential / Power Input 1	3509	Sensor Supply 1
430	Engine Starter Solenoid Voltage	3510	Sensor Supply 2
444	Battery Potential / Power Input 2	3511	Sensor Supply 3
677	Engine Starter Motor Relay	3512	Sensor Supply 4
1204	Electrical Load	3513	Sensor Supply 5
1321	Engine Starter Solenoid Lockout Relay Driver Circuit	3514	Sensor Supply 6
1795	Alternator Current (High Range/Resolution)	3597	ECU Power Output 1
1800	Battery 1 Temperature	3598	ECU Power Output 2
1801	Battery 2 Temperature		

(R) FIGURE PGN65104_B – BATTERY INFORMATION LEGEND